Hack-Nation

In collaboration with MIT Club of Northern California and MIT Club of Germany

CHALLENGE

05

Foundation Models for Women's Hormonal Health

Building the AI Infrastructure for the Next Generation of Women's Health

POWERED BY

Hack-Nation

ENABLED BY

OpenAI

Goals and Motivation

Meet Sarah

Sarah is 34. For years she has struggled with fatigue, irregular cycles, migraines, and brain fog — seeing multiple physicians without clear answers. Each appointment captures only a snapshot, yet hormones shift continuously with sleep, stress, nutrition, and age.

Women represent more than 50% of the global population, yet female physiology remains one of the least represented domains in AI and biomedical research. Hormones shape brain function, cardiovascular health, and metabolism — but the AI infrastructure to understand these interactions largely does not exist.

Conditions such as PCOS, endometriosis, and menopause-related disease often take years to diagnose, and heart attacks in women are more likely to go unrecognized. Closing the women's health gap could unlock more than $1 trillion in annual global economic impact.

Every AI breakthrough has been built on shared infrastructure — ImageNet, AlphaFold, and the open datasets behind today's language models. Women's hormonal health deserves the same foundation.

Current Challenges

NO SHARED BENCHMARK	Researchers lack standardized multimodal datasets combining hormone measurements, wearables, laboratory data, symptoms, and longitudinal physiological signals.
FRAGMENTED INFRASTRUCTURE	Data, models, and benchmarks are scattered across institutions, making approaches difficult to compare and research slow to compound.
DYNAMIC BIOLOGY, STATIC CARE	Clinical decisions still rely on occasional laboratory tests rather than continuous understanding of hormonal physiology over time.

Your Challenge

Help build the world's first open AI infrastructure for women's hormonal health. Rather than attempting one large foundation model in a weekend, contribute a single reusable building block — a dataset, benchmark, model, or application — that future researchers can immediately build on.

Most importantly: publish your datasets, benchmarks, models, and code under an open license whenever possible, and choose the layer where your team can contribute most.

01 Data & Benchmark Infrastructure

Curate, integrate, and publish the data foundations that let researchers train, evaluate, and compare AI models for women's hormonal health.

- A standardized multimodal dataset combining public wearables, laboratory measurements, imaging, symptoms, and longitudinal signals.
- Benchmark datasets with documented train / validation / test splits and transparent evaluation methodology.
- Focused prediction tasks such as hormone level, ovulation, menopause stage, or disease-risk prediction.

Goal: Leave behind an open dataset or benchmark that future researchers can use immediately.

Build One Reusable Layer

02 AI Model Infrastructure

Train a focused model that contributes toward future foundation models — optimizing for reproducibility, scientific rigor, and explainability over sheer size.

- Hormone-level or hormonal-state prediction, including biological signatures of early or late menopause onset.
- Explainable AI that integrates multiple data sources and interprets findings in the context of hormonal variability.
- Prediction of clinical trials where women's hormonal variability may affect efficacy or safety.

Goal: Build a reusable model that researchers can reproduce, extend, and combine into larger systems.

03 Application Infrastructure

Demonstrate how better AI infrastructure improves women's lives by building on reusable datasets and models to solve one clearly defined problem.

- A regulator-conscious pathway for volunteers to contribute de-identified EHR or longitudinal health data to research.
- AI symptom tracking, digital hormone journals, or voice-based health logging.
- Personalized hormone insights or digital twins of female physiology that help address gaps in access and representation.

Goal: Translate foundational AI into meaningful, measurable impact for women.

Open Science Requirement

Every major AI breakthrough has been accelerated through open science. Whenever possible, publish your datasets, benchmarks, source code, model checkpoints, documentation, and evaluation pipelines.

The objective is not simply a prototype — it is a reusable scientific asset that accelerates research long after the hackathon ends.

OpenAI Tools & Credits

We recommend exploring OpenAI's models for this challenge. Hack-Nation is providing $50 in free OpenAI API credits per team, available on a first-come, first-served basis.

- Explore multimodal approaches that combine data across domains, including OpenAI's multimodal models and image generation (gpt-image-2) alongside other modality-specific models.
- Strong entries demonstrate multimodal integration, clear user value, technical quality, creativity, responsible design, and a convincing demo.

Data Sources & Hints

Ground your project in openly available datasets whenever possible. Teams are encouraged to integrate multiple datasets or responsibly collect new longitudinal data while complying with applicable privacy, ethical, and licensing requirements.

Dataset	Description	Link
mcPHASES (PhysioNet)	Fitbit, continuous glucose monitoring, hormone measurements, menstrual-cycle data, sleep, and symptoms.	physionet.org/content/mcphases
NHANES (CDC)	Reproductive health, thyroid hormones, laboratory data, nutrition, and demographics.	wwwn.cdc.gov/nchs/nhanes

Other datasets may be available, and expanding this list is encouraged.

Deliverables

Choose one specific hormone-related problem and demonstrate how your solution advances research and improves women's health.

- Working prototype and source code
- Technical documentation and dataset description
- Benchmark methodology and a short demonstration video

What Makes a Strong Submission

Strong Submissions…	Weak Submissions…
— Publish reusable datasets, benchmarks, model checkpoints, and evaluation pipelines under an open license.	— Build an isolated application with no reusable dataset, benchmark, or model contribution.
— Solve one clearly defined prediction or infrastructure problem exceptionally well, with transparent methods and evidence.	— Depend on undocumented proprietary data or hide core assumptions, preprocessing, and evaluation choices.
— Share reproducible code and documentation so the research community can extend the work after the hackathon.	— Make unsupported medical or diagnostic claims, or present a polished interface without scientific validation.

Success Criteria

01 Women's Health Impact: How significantly could this work improve women's health, and how many women could ultimately benefit?

02 Technical Excellence: How innovative, rigorous, reproducible, and scalable is the proposed dataset, benchmark, model, or application?

03 Foundation Value: Does the project leave behind reusable infrastructure that accelerates future research?

Measure the Impact

REACH	QUALITY OF LIFE
How many women could ultimately benefit from the dataset, benchmark, model, or application?	How meaningfully could the solution improve health outcomes, access, understanding, or daily well-being?

Why This Matters

The next generation of AI for women's health will not be built by a single company. It will emerge from shared datasets, reproducible benchmarks, reusable AI models, and open scientific collaboration.

Women's hormonal health remains underrepresented in biomedical research and AI infrastructure. Shared scientific assets can help researchers compare methods, reproduce results, and compound progress across institutions instead of rebuilding the same foundations in isolation.

This challenge is about creating scientific infrastructure that accelerates research and helps unlock a trillion-dollar opportunity to improve health outcomes and quality of life for women worldwide.